Yiwei Li

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3,300+ citations | First-author publications at **ICLR, ECCV, and NeurIPS**

Education

University of Georgia Ph.D. in Computer Science — Aug 2023 – Dec 2026 (expected)

Advisor: Tianming Liu. Research collaborator: Xiang Li, Massachusetts General Hospital / Harvard Medical School.

Multimodal reasoning, RL post-training, controllable video generation, world models, and representation learning.

Southeast University M.S. in Computer Science and Engineering — Sep 2020 – Jun 2023

Southeast University B.Eng., Chien-Shiung Wu Honors College — 2016 – 2020

Experience

Research Scientist Intern @ United Imaging Intelligence — Boston, MA | May – Aug 2025

1. **Multimodal reasoning and RL post-training for visual grounding** — Paper #1 (ECCV)

- Formulated **RAU**, reference-guided multimodal reasoning that transfers an annotated example to an unlabeled target, unifying visual identification, box localization, and segmentation without task-specific target annotation.
- Coupled **DINOv2 reference retrieval, Qwen2.5-VL-7B, and SAM2** through a learned segmentation-token-to-memory adapter; used relational reasoning and optimal-transport box matching for spatial grounding.
- Developed **SFT + GRPO** with task-specific accuracy and format rewards; trained a 650K-example VQA setting with **LoRA and 4-bit adaptation on 8 A100 GPUs**.
- Improved mean **Dice from 0.24 to 0.71** and gIoU from 0.14 to 0.50 over the reference-conditioned SAM2 SFT baseline across four datasets, including two OOD datasets. First author, **ECCV 2026**.

Research Intern @ MGH / Harvard Medical School — Remote | May 2024 – Present

2. **Controllable video generation from time-series signals** — Paper #2 (ICLR)

- Co-developed **EchoPulse**, a signal-conditioned video model that learns the relationship between physiological time series and visual dynamics, enabling controllable synthesis from ECG inputs.
- Designed a **spatiotemporal VQ-VAE tokenizer and masked visual token model** with iterative parallel decoding; incorporated temporal conditioning, classifier-free guidance, and autoregressive extension.
- Achieved **6.4 s generation versus 146 s** for EchoDiffusion (~23× faster), while reducing ejection-fraction MAE from **4.81 to 2.51**, in the released project benchmark.
- Evaluated generation across **three public/private datasets** and released training code, model weights, inference notebooks, and wearable-ECG demonstrations. First author, **ICLR 2025**.

Ph.D. Research @ University of Georgia — Athens, GA | Aug 2023 – Present

3. **World models and representation learning from expert behavior** — Paper #3 (NeurIPS)

- Developed **GazeWorld**, treating images as perceptual environments and expert gaze as trajectories; combined causal next-fixation latent prediction with spatial completion of unvisited image regions.
- Learned transferable features using an online encoder and **EMA target encoder**; expert behavior supervises pre-training, while downstream inference requires **images alone**.
- Evaluated frozen representations across **three external datasets at 1%, 10%, and 100% labels** to test transfer and data efficiency; ablated trajectory order, predictive objectives, and backbone initialization.
- Improved **ScanMatch by over 16%** and reduced scanpath edit distance by **over 22%** versus LogitGaze-Med (CheX), using a generic decoder on frozen features. First author, **NeurIPS 2026**.

Research Scientist Intern @ United Imaging Intelligence — Boston, MA | May – Aug 2026

- Built a **DINOv3-based video world model** for temporal feature prediction; developed cross-projection self-distillation with Gram anchoring and compared contrastive, masked-prediction, and EMA-teacher objectives.

Experience (continued)

Additional Ph.D. Research @ University of Georgia — Aug 2023 – Present

- Developed **Thinking with Gaze**: dedicated gaze tokens and staged LoRA adaptation turn expert visual search into VLM reasoning supervision; improved MIMIC-EYE AUROC from **87.60 to 90.17** with transfer to three external benchmarks.
- Ran JEPA-style representation-learning experiments on **16 H200 GPUs**, studying masking strategies, distributed training, objective ablations, and downstream transfer.
- Developed **ALDM-Grasping**, combining diffusion with adversarial layout supervision for sim-to-real transfer; achieved **75% / 65%** grasp success on plain / complex backgrounds without real-image training.

Software Engineer Intern @ Huawei — Jun – Sep 2020

Worked on AI-assisted telecommunications algorithms for Massive-MIMO transmitter systems.

Selected Lead-Author Publications

Full publication list available on Google Scholar.

1. **RAU: Reference-based Anatomical Understanding with Vision Language Models** **[paper]**
   Yiwei Li, Yikang Liu, Jiaqi Guo, Lin Zhao, et al. | **ECCV 2026**
   Retrieval-guided VLM reasoning and RL post-training connect reference examples to transferable pixel-level grounding.
2. **ECHOPulse: ECG Controlled Echocardiograms Video Generation** **[paper]**
   Yiwei Li, Sekeun Kim, Zihao Wu, Hanqi Jiang, et al. | **ICLR 2025**
   Time-series-conditioned masked-token modeling enables controllable, efficient video generation.
3. **A World Model of Radiologist Reading for Medical Image Representation Learning** **[paper]**
   Yiwei Li, Zihao Wu, Huaqin Zhao, Yifan Zhou, Chao Cao, Dajiang Zhu, Tianming Liu, Lin Zhao | **NeurIPS 2026**
   Expert viewing trajectories supervise latent prediction to learn image representations without gaze at inference.

Open Source Contributions

EchoPulse Video-generation training/inference, pretrained weights, and wearable-input demonstration notebooks. [code]

AD-AutoGPT LangChain agent for goal decomposition, tool invocation, document processing, and analysis. [code]

NRRS / Vaa3D C++ neuron-retracing and morphology-refinement implementation in the Vaa3D tool ecosystem. [code]

Technical Skills

Programming & tools: Python, C/C++, Linux, Git; PyTorch, Hugging Face Accelerate, NumPy, torchvision, einops, Weights & Biases, LangChain.

Training & adaptation: SFT, GRPO, reward design, LoRA, 4-bit adaptation, FP16/BF16, multi-GPU training, gradient accumulation/clipping, checkpointing; 7B VLMs and 70B LLM adaptation.

Models & evaluation: Qwen2.5-VL, Llama 3, SAM2, DINOv2/v3, JEPA, VQ-VAE, MaskGIT-style generation, diffusion; FID/FVD, Dice/IoU, AUROC, linear probing, OOD evaluation.

Other Relevant Publications

MEDQUA, ISBI 2026 (Oral; first author) · **TTT-Unet**, MIDL 2026 (Oral) · **SAMed-2**, MICCAI 2025 · **HARP**, ICRA 2025 · **EGMA**, NeurIPS 2024 · **AD-AutoGPT**, PLOS Global Public Health 2025 (first author).

Service, Awards, and Teaching

- **Area Chair:** GenAI4Health Workshop at NeurIPS 2026. **Research Award:** UGA School of Computing, 2026.
- **Reviewing (75+ reviews):** ICLR, NeurIPS, ICML, AAAI, CVPR, ICCV, ECCV, MICCAI; IEEE TMI, Medical Image Analysis, Pattern Recognition. **Graduate Teaching Assistant:** UGA, Fall 2025 / Spring 2026.